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LEARNING HUB – ONLINE LEARNING PLATFORM

30-Day Development Plan | 4 Weeks | 6 Modules

Tech Stack: Python Django + React.js + SQLite/MySQL

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OVERVIEW

Total Days : 30

Total Weeks : 4

Modules : 6

Backend : Python Django + Django REST Framework

Frontend : React.js + Axios + React Router

Database : SQLite (dev) → MySQL (production)

Auth : JWT (djanoorestframework-simplejwt)

Deployment : Render (backend) + Vercel (frontend)

MODULE LIST:

Module 1 – User Management

Module 2 – Course Management

Module 3 – Learning Delivery

Module 4 – Assessment (Quizzes & Assignments)

Module 5 – Student Dashboard

Module 6 – Feedback & Reviews

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WEEK 1 – SETUP & USER AUTHENTICATION (Days 1-7)

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DAY 1 – Project Environment Setup

Goal : Get Django running and admin panel accessible

Tasks:

[x] Install Python (3.11+) and VS Code extensions

- Python by Microsoft
- Pylance by Microsoft
- Django by Baptiste Darthenay

[x] Create project folder: D:\Online_lear_plt

[x] Create subfolders: backend/ and frontend/

[x] Create and activate virtual environment

Command: python -m venv venv

Activate (Windows): venv\Scripts\activate

```
48 [x] Install all required packages:
49     pip install django djangorestframework
50         djangorestframework-simplejwt
51         django-cors-headers Pillow python-dotenv
52 [x] Save packages: pip freeze > requirements.txt
53 [x] Create Django project: django-admin startproject learning_hub .
54 [x] Run initial migrations: python manage.py migrate
55 [x] Create superuser: python manage.py createsuperuser
56 [x] Start server: python manage.py runserver
```

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57
58 End of Day Goal:
59     → Django admin panel accessible at http://127.0.0.1:8000/admin
60
```

```
61 -----
62 DAY 2 – Django Apps & Settings Configuration
63 -----
```

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64 Goal      : Create all 4 apps and configure settings.py fully
```

```
65
```

```
66 Tasks:
```

```
67 [x] Create 4 Django apps:
68     python manage.py startapp users
69     python manage.py startapp courses
70     python manage.py startapp enrollments
71     python manage.py startapp assessments
72 [x] Add apps to INSTALLED_APPS in settings.py:
73     'rest_framework', 'corsheaders',
74     'users', 'courses', 'enrollments', 'assessments'
75 [x] Add CorsMiddleware at top of MIDDLEWARE
76 [x] Configure CORS_ALLOWED_ORIGINS = ["http://localhost:3000"]
77 [x] Set REST_FRAMEWORK default auth to JWTAuthentication
78 [x] Set AUTH_USER_MODEL = 'users.CustomUser'
79 [x] Set MEDIA_URL and MEDIA_ROOT for file uploads
80 [x] Create .env file with SECRET_KEY and DEBUG=True
81 [x] Load .env in settings.py using python-dotenv
```

```
82
83 End of Day Goal:
84     → All apps created, settings fully configured, server still running
85
```

```
86 -----
87 DAY 3 – Custom User Model (MODULE 1 START)
88 -----
```

```
89 Module      : Module 1 – User Management
90 Goal        : Build CustomUser model with 3 roles
```

```
91
```

```
92 Tasks:
```

```
93 [x] Write CustomUser model in users/models.py
94     - Extend AbstractUser
```

```
95         - Add role field with choices: admin, instructor, student
96         - Add bio (TextField) and profile_picture (ImageField)
97     [x] Write UserSerializer in users/serializers.py
98     [x] Register CustomUser in users/admin.py using CustomUserAdmin
99         - Add list_display: username, email, role, is_staff
100    [x] Run: python manage.py makemigrations users
101    [x] Run: python manage.py migrate
102    [x] Verify role column appears in admin panel
103
104    End of Day Goal:
105        → CustomUser model visible in admin with Role column
106
107    -----
108    DAY 4 – Register & Login API
109    -----
110    Module      : Module 1 – User Management
111    Goal        : Build working register and login endpoints with JWT
112
113    Tasks:
114        [x] Write RegisterSerializer in users/serializers.py
115            - Fields: username, email, password, role
116            - Use create_user() to hash password
117        [x] Write RegisterView in users/views.py
118            - POST /api/users/register/
119            - permission_classes = [AllowAny]
120        [x] Write LoginView in users/views.py
121            - POST /api/users/login/
122            - Returns JWT access + refresh tokens + role
123        [x] Write ProfileView in users/views.py
124            - GET /api/users/profile/
125            - permission_classes = [IsAuthenticated]
126        [x] Create users/urls.py with routes:
127            - register/, login/, profile/
128        [x] Include users.urls in learning_hub/urls.py
129        [x] Test register via browser at:
130            http://127.0.0.1:8000/api/users/register/
131
132    End of Day Goal:
133        → Register and login return real JWT access tokens
134
135    -----
136    DAY 5 – Role-Based Permissions
137    -----
138    Module      : Module 1 – User Management
139    Goal        : Protect routes based on user role
140
141    Tasks:
```

```
142 [ ] Create users/permissions.py with custom permission classes:
143     - IsAdmin: user.role == 'admin'
144     - IsInstructor: user.role == 'instructor'
145     - IsStudent: user.role == 'student'
146 [ ] Add profile update view (PATCH /api/users/profile/update/)
147 [ ] Apply permission classes to views
148 [ ] Test: access protected route without token → 401 Unauthorized
149 [ ] Test: login as student → cannot access instructor route
150
151 End of Day Goal:
152     → Role-based access enforced on all protected API routes
153
154 -----
155 DAY 6 – React Frontend Setup + Auth UI
156 -----
157 Module      : Module 1 – User Management
158 Goal        : Build Login and Register screens in React
159
160 Tasks:
161 [ ] Install Node.js from nodejs.org
162 [ ] Run: npx create-react-app . inside frontend/
163 [ ] Install packages: npm install axios react-router-dom
164 [ ] Set up React Router in App.js with routes:
165     /, /login, /register, /dashboard
166 [ ] Create src/services/api.js with axios base URL:
167     http://localhost:8000
168 [ ] Build LoginPage component
169     - Fields: username, password
170     - POST to /api/users/login/
171 [ ] Build RegisterPage component
172     - Fields: username, email, password, role dropdown
173     - POST to /api/users/register/
174 [ ] Start React: npm start
175 [ ] Verify both pages open at http://localhost:3000
176
177 End of Day Goal:
178     → Login and Register pages visible in browser
179
180 -----
181 DAY 7 – Connect Frontend ↔ Backend + Auth State
182 -----
183 Module      : Module 1 – User Management
184 Goal        : Wire React to Django API, store JWT, protect routes
185
186 Tasks:
187 [ ] Connect LoginPage to POST /api/users/login/ via axios
188 [ ] Save JWT access token to localStorage on login
```

```
189 [ ] Create AuthContext using React Context API
190     - Store: token, user, role
191     - Functions: login(), logout()
192 [ ] Create PrivateRoute component
193     - Redirects to /login if no token found
194 [ ] Protect /dashboard route with PrivateRoute
195 [ ] Test full flow:
196     Register → Login → Token saved → Redirect to dashboard
197 [ ] Commit all Week 1 work to GitHub
198
199 End of Day Goal:
200     → Full authentication flow working end-to-end ✓ MODULE 1 COMPLETE
201
202 =====
203 WEEK 2 – COURSES & LEARNING DELIVERY (Days 8-16)
204 =====
205
206 -----
207 DAY 8 – Course Model & Database
208 -----
209 Module      : Module 2 – Course Management
210 Goal        : Build Course, Lesson, and Material database models
211
212 Tasks:
213 [ ] Write Course model in courses/models.py:
214     - title, description, instructor (FK → User)
215     - category, thumbnail (ImageField), created_at
216 [ ] Write Lesson model:
217     - title, course (FK), video_file, order_number, duration
218 [ ] Write CourseMaterial model:
219     - lesson (FK), file, material_type (pdf/doc/video)
220 [ ] Run: python manage.py makemigrations courses
221 [ ] Run: python manage.py migrate
222 [ ] Register all models in courses/admin.py
223 [ ] Add sample course via admin panel
224
225 End of Day Goal:
226     → Course tables in database, visible and manageable from admin
227
228 -----
229 DAY 9 – Course CRUD API
230 -----
231
232 Module      : Module 2 – Course Management
233 Goal        : Build full Create, Read, Update, Delete API for courses
234
235 Tasks:
236 [ ] Write CourseSerializer and LessonSerializer
```

```
236 [ ] Write CourseListCreateView:
237     - GET: all courses (public)
238     - POST: instructors only
239 [ ] Write CourseDetailView:
240     - GET: course details + lesson list
241     - PUT/DELETE: instructor who owns course only
242 [ ] Write LessonCreateView (instructor only)
243 [ ] Set up courses/urls.py and include in root urls.py
244 [ ] Add file upload support for video_file and thumbnail
245 [ ] Test: create course via API, fetch via GET
246
247 End of Day Goal:
248     → Full course CRUD API working and tested
249
250 -----
251 DAY 10 – Enrollment Model & API
252 -----
253 Module      : Module 3 – Learning Delivery
254 Goal        : Students can enroll in courses and track progress
255
256 Tasks:
257 [ ] Write Enrollment model in enrollments/models.py:
258     - student (FK User), course (FK Course)
259     - enrolled_at, is_completed
260 [ ] Write Progress model:
261     - enrollment (FK), lesson (FK)
262     - is_completed (bool), completed_at
263 [ ] Write EnrollView (POST /api/enrollments/enroll/)
264     - Students only, prevent duplicate enrollment
265 [ ] Write MyCoursesView (GET enrolled courses for logged-in student)
266 [ ] Write MarkLessonCompleteView (PATCH)
267     - Marks lesson done + calculates progress %
268 [ ] Run migrations and test enrollment via API
269
270 End of Day Goal:
271     → Students can enroll and lesson progress is tracked
272
273 -----
274 DAY 11 – Course Listing Page (React)
275 -----
276 Module      : Module 2 – Course Management (Frontend)
277 Goal        : Students can browse and enroll in courses
278
279 Tasks:
280 [ ] Build CoursesPage:
281     - Fetches GET /api/courses/
282     - Renders grid of CourseCard components
```

- [] Build CourseCard component:
 - Shows thumbnail, title, instructor name, enroll button
- [] Add search bar to filter courses by title
- [] Build CourseDetailPage:
 - Shows description, instructor, lesson list
- [] Wire "Enroll" button to POST /api/enrollments/enroll/
- [] Show "Already enrolled" if student is already enrolled

End of Day Goal:
→ Students can browse all courses and click to enroll

DAY 12 – Video Player & Lesson Viewer

Module : Module 3 – Learning Delivery (Frontend)
Goal : Students can watch video lectures inside the app

Tasks:

- [] Build LessonPage layout:
 - Left sidebar: lesson list with checkmarks
 - Right panel: HTML5 video player
- [] Play uploaded video files from /media/ folder
- [] Add "Mark as Complete" button
 - Calls PATCH /api/enrollments/complete-lesson/
- [] Show progress bar at top (% of lessons completed)
- [] List downloadable materials below the video
- [] Auto-navigate to next lesson after completion

End of Day Goal:
→ Students can watch videos and track lesson progress

DAY 13 – Instructor Dashboard

Module : Module 2 – Course Management (Frontend)
Goal : Instructors can create and manage their courses

Tasks:

- [] Build InstructorDashboard page:
 - Shows all courses created by this instructor
 - Displays enrolled student count per course
- [] Build CreateCoursePage form:
 - Fields: title, description, category, thumbnail upload
 - POST to /api/courses/
- [] Build AddLessonPage form:
 - Fields: title, video upload, order number
- [] Add Edit and Delete buttons on course cards

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330         - Instructor can only edit/delete their own courses
331     [ ] Show success/error messages after actions
332
333 End of Day Goal:
334     → Instructors can fully create, edit and delete their courses
335
336 -----
337 DAY 14 – Student Dashboard
338 -----
339 Module      : Module 5 – Student Dashboard
340 Goal        : Students have a personal learning overview page
341
342 Tasks:
343     [ ] Build StudentDashboard page:
344         - Fetch enrolled courses from /api/enrollments/my-courses/
345         - Show each course with a progress bar
346     [ ] Add "Continue Learning" button:
347         - Links to last incomplete lesson
348     [ ] Show completed courses with a green "Completed" badge
349     [ ] Add profile section:
350         - Student name, role, profile picture
351         - Edit profile button
352     [ ] Show total courses enrolled and completed as stats
353
354 End of Day Goal:
355     → Student dashboard fully functional with progress display ✓ MODULE 5 COMPLETE
356
357 -----
358 DAYS 15-16 – Buffer & Review Days
359 -----
360 Goal        : Fix bugs, polish Week 2 work, commit everything
361
362 Tasks:
363     [ ] Re-test all auth flows (register, login, token expiry)
364     [ ] Re-test enrollment and progress tracking
365     [ ] Fix any broken API calls between React and Django
366     [ ] Make all pages responsive on mobile (check with dev tools)
367     [ ] Clean up code: remove console.log, add comments
368     [ ] Make sure video playback works correctly
369     [ ] Commit all Week 1 + Week 2 work to GitHub
370         - Use clear commit messages
371
372 End of Day Goal:
373     → Weeks 1-2 stable, committed to GitHub ✓ MODULES 2 & 3 COMPLETE
374
375 =====
376 WEEK 3 – ASSESSMENTS & FEEDBACK (Days 17-23)
```



```
377 =====
378
379 -----
380 DAY 17 – Quiz Model & Database
381 -----
382 Module      : Module 4 – Assessment
383 Goal        : Build Quiz, Question, Option database models
384
385 Tasks:
386     [ ] Write Quiz model in assessments/models.py:
387         - title, course (FK), total_marks, duration_minutes
388     [ ] Write Question model:
389         - quiz (FK), question_text, marks
390     [ ] Write Option model:
391         - question (FK), option_text, is_correct (boolean)
392     [ ] Write QuizSubmission model:
393         - student (FK), quiz (FK), score, submitted_at
394     [ ] Run: python manage.py makemigrations assessments
395     [ ] Run: python manage.py migrate
396     [ ] Register all models in assessments/admin.py
397     [ ] Add a sample quiz with questions via admin panel
398
399 End of Day Goal:
400     → Quiz tables ready, sample quiz visible in admin panel
401
402 -----
403 DAY 18 – Quiz Submission & Auto-Scoring
404 -----
405 Module      : Module 4 – Assessment
406 Goal        : Students submit quiz, score is automatically calculated
407
408 Tasks:
409     [ ] Write SubmitQuizView:
410         - POST /api/assessments/quiz/submit/
411         - Accepts: {quiz_id, answers: [{question_id, option_id}]}
412         - Checks is_correct for each answer
413         - Calculates and saves total score
414     [ ] Write QuizResultView:
415         - Returns score and correct answers after submission
416     [ ] Write StudentGradesView:
417         - Lists all quiz scores for the logged-in student
418     [ ] Prevent resubmission (check if submission exists)
419     [ ] Test: submit quiz via API, verify score is correct
420
421 End of Day Goal:
422     → Auto-scoring works correctly, grades saved to database
423
```

```
424 -----
425 DAY 19 – Assignment Submission System
426 -----
427 Module      : Module 4 – Assessment
428 Goal        : Students upload files, instructors grade them
429
430 Tasks:
431   [ ] Write Assignment model:
432       - title, course (FK), due_date, description
433   [ ] Write AssignmentSubmission model:
434       - student (FK), assignment (FK)
435       - submitted_file (FileField), grade, feedback
436   [ ] Write SubmitAssignmentView:
437       - POST: accepts file upload from student
438   [ ] Write GradeAssignmentView (instructor only):
439       - PATCH: instructor sets grade (0-100) + written feedback
440   [ ] Write MySubmissionsView:
441       - GET: student sees their submissions + grades
442   [ ] Run migrations and test file upload via API
443
444 End of Day Goal:
445   → Students can upload assignments, instructors can grade them
446
447 -----
448 DAY 20 – Quiz UI in React
449 -----
450 Module      : Module 4 – Assessment (Frontend)
451 Goal        : Students can take quizzes in the browser
452
453 Tasks:
454   [ ] Build QuizPage:
455       - Fetches questions from /api/assessments/quiz/{id}/
456       - Shows one question at a time with radio button options
457   [ ] Track selected answers in React useState
458   [ ] Add countdown timer using useEffect
459       - Auto-submits when timer reaches 0
460   [ ] On submit: POST answers to /api/assessments/quiz/submit/
461   [ ] Build QuizResultPage:
462       - Shows final score (e.g. 8/10)
463       - Highlights correct answers in green, wrong in red
464
465 End of Day Goal:
466   → Students can take timed quizzes and see their results
467
468 -----
469 DAY 21 – Feedback & Reviews Module
470 -----
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```
471 Module      : Module 6 – Feedback & Reviews
472 Goal        : Course ratings, reviews and instructor announcements
473
474 Tasks:
475   [ ] Write Review model:
476       - student (FK), course (FK), rating (1-5)
477       - comment (TextField), created_at
478   [ ] Write Announcement model:
479       - instructor (FK), course (FK)
480       - title, body, created_at
481   [ ] Write AddReviewView (students only, one per course)
482   [ ] Write CourseReviewsView (GET all reviews for a course)
483   [ ] Write PostAnnouncementView (instructors only)
484   [ ] Write CourseAnnouncementsView (GET announcements)
485   [ ] Calculate average star rating per course
486   [ ] Display average rating on CourseCard in React
487
488 End of Day Goal:
489   → Reviews, ratings and announcements working ✓ MODULE 6 COMPLETE
490
491 -----
492 DAYS 22-23 – Assignment UI + Grades Page
493 -----
494 Module      : Module 4 – Assessment (Frontend)
495 Goal        : Full assignment and grading flow in the browser
496
497 Tasks:
498   [ ] Build AssignmentPage:
499       - Shows assignment title, description, due date
500       - File upload form (PDF/DOC)
501       - Submit button → POST to /api/assessments/assignment/submit/
502   [ ] Build GradesPage:
503       - Table of all quiz scores and assignment grades
504       - Shows: subject, score, date, instructor feedback
505   [ ] Build InstructorGradingPage:
506       - Lists all student submissions for an assignment
507       - Grade input (0-100) + feedback text box per student
508       - Save button → PATCH to /api/assessments/grade/
509   [ ] Test full cycle:
510       Student submits → Instructor grades → Student sees grade
511   [ ] Commit all Week 3 work to GitHub
512
513 End of Day Goal:
514   → Full assessment cycle working end-to-end ✓ MODULE 4 COMPLETE
515
516 =====
517 WEEK 4 – TESTING, POLISH & DEPLOYMENT (Days 24-30)
```

```
518 =====
519
520 -----
521 DAY 24 – Full Application Testing
522 -----
523 Goal      : Test every user flow from start to finish
524
525 Tasks:
526   [ ] Test Student flow:
527       Register → Login → Browse courses → Enroll
528       → Watch video → Mark complete → Take quiz → View grade
529   [ ] Test Instructor flow:
530       Login → Create course → Add lessons → Post assignment
531       → View submissions → Grade student work
532   [ ] Test Admin flow:
533       Login → View all users → Manage courses → View enrollments
534   [ ] Test edge cases:
535       - Wrong password → error message shown
536       - Empty form submit → validation errors shown
537       - Duplicate enrollment → blocked with message
538       - Quiz resubmission → blocked
539       - Assignment past due date → show warning
540   [ ] Write down all bugs found in a bug list
541
542 End of Day Goal:
543   → Complete bug list documented, ready to fix
544
545 -----
546 DAYS 25-26 – Bug Fixing & UI Polish
547 -----
548 Goal      : Fix all bugs and improve the look and feel
549
550 Tasks:
551   [ ] Fix all bugs found on Day 24
552   [ ] Add loading spinner on all API calls:
553       - Use React useState(loading) pattern
554       - Show spinner while waiting for response
555   [ ] Add proper error messages throughout:
556       - "Invalid credentials" on failed login
557       - "Already enrolled in this course"
558       - "Quiz already submitted"
559   [ ] Make all pages mobile responsive:
560       - Check with browser developer tools (F12)
561       - Fix any broken layouts at small screen sizes
562   [ ] Build a Navbar with role-aware links:
563       - Student: Home, My Courses, Grades
564       - Instructor: Dashboard, Create Course
```

```
565         - Admin: Users, All Courses
566     [ ] Add logout button that clears token and redirects to login
567
568 End of Day Goal:
569     → App is polished, responsive, no major bugs remaining
570
571 -----
572 DAY 27 – Prepare for Deployment
573 -----
574 Goal      : Configure the app for production environment
575
576 Tasks:
577     [ ] Set DEBUG=False in .env
578     [ ] Add ALLOWED_HOSTS in settings.py:
579         ALLOWED_HOSTS = ['localhost', '127.0.0.1', '.render.com']
580     [ ] Run: python manage.py collectstatic
581     [ ] Install gunicorn: pip install gunicorn
582     [ ] Update requirements.txt: pip freeze > requirements.txt
583     [ ] Create .gitignore file in root:
584         venv/, .env, __pycache__/, *.pyc
585         node_modules/, build/, .DS_Store
586     [ ] Final push of all code to GitHub
587     [ ] Make sure GitHub repo has both backend/ and frontend/ folders
588
589 End of Day Goal:
590     → All code pushed to GitHub, ready for live deployment
591
592 -----
593 DAY 28 – Deploy Backend on Render
594 -----
595 Goal      : Put Django backend live on the internet
596
597 Tasks:
598     [ ] Sign up at render.com
599     [ ] New Web Service → Connect your GitHub repository
600     [ ] Set Build Command:
601         pip install -r requirements.txt
602     [ ] Set Start Command:
603         gunicorn learning_hub.wsgi:application
604     [ ] Add Environment Variables on Render dashboard:
605         SECRET_KEY = your-secret-key
606         DEBUG = False
607         ALLOWED_HOSTS = .render.com
608     [ ] Wait for deployment to complete
609     [ ] Test live backend:
610         https://your-app.onrender.com/admin
611         https://your-app.onrender.com/api/users/register/
```

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612     [ ] Create superuser on Render:
613         python manage.py createsuperuser (via Render shell)
614
615 End of Day Goal:
616     → Django backend is live and accessible at a real URL
617
618 -----
619 DAY 29 – Deploy Frontend on Vercel
620 -----
621 Goal      : Put React frontend live on the internet
622
623 Tasks:
624     [ ] Update src/services/api.js:
625         Change base URL from http://localhost:8000
626         to your Render backend URL
627     [ ] Build React app: npm run build
628     [ ] Sign up at vercel.com
629     [ ] Import Project → Connect frontend GitHub folder
630     [ ] Vercel auto-detects React → Click Deploy
631     [ ] Wait for deployment (usually under 2 minutes)
632     [ ] Test the full live application:
633         - Register a new account
634         - Login and check role redirect
635         - Enroll in a course
636         - Take a quiz
637         - All done on live URLs!
638
639 End of Day Goal:
640     → Full app live – frontend on Vercel + backend on Render
641
642 -----
643 DAY 30 – Final Review & Documentation
644 -----
645 Goal      : Write documentation and prepare final deliverable
646
647 Tasks:
648     [ ] Write README.md file with:
649         - Project title and description
650         - Tech stack list
651         - How to run locally (step-by-step)
652         - Live demo link (Vercel URL)
653         - Admin login details (for reviewer)
654     [ ] Do a complete walkthrough of the live app:
655         - Login as Student → test all student features
656         - Login as Instructor → test course creation
657         - Login as Admin → check admin panel
658     [ ] Take screenshots of all key screens:
```

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659         - Login page, Student dashboard, Course page
660         - Quiz page, Instructor dashboard, Admin panel
661     [ ] Prepare project report / synopsis update
662     [ ] Final GitHub commit with message:
663         "Final submission - Learning Hub v1.0"
664
665 End of Day Goal:
666     → Project complete and ready to submit ✓ ALL 6 MODULES COMPLETE
667
668 =====
669 MODULE COMPLETION SUMMARY
670 =====
671
672 Module 1 – User Management           → Completed: Day 7
673 Module 2 – Course Management        → Completed: Day 15
674 Module 3 – Learning Delivery        → Completed: Day 16
675 Module 4 – Assessment               → Completed: Day 23
676 Module 5 – Student Dashboard        → Completed: Day 14
677 Module 6 – Feedback & Reviews       → Completed: Day 21
678
679 =====
680 API ENDPOINTS SUMMARY
681 =====
682
683 USER ENDPOINTS:
684 POST    /api/users/register/         → Register new user
685 POST    /api/users/login/            → Login + get JWT token
686 GET     /api/users/profile/          → Get logged-in user profile
687 PATCH   /api/users/profile/update/   → Update profile
688
689 COURSE ENDPOINTS:
690 GET     /api/courses/                → List all courses
691 POST    /api/courses/                → Create course (instructor)
692 GET     /api/courses/{id}/           → Course detail + lessons
693 PUT     /api/courses/{id}/           → Update course (owner only)
694 DELETE  /api/courses/{id}/           → Delete course (owner only)
695
696 ENROLLMENT ENDPOINTS:
697 POST    /api/enrollments/enroll/     → Enroll in a course
698 GET     /api/enrollments/my-courses/ → Student's enrolled courses
699 PATCH   /api/enrollments/complete-lesson/ → Mark lesson complete
700
701 ASSESSMENT ENDPOINTS:
702 GET     /api/assessments/quiz/{id}/  → Get quiz questions
703 POST    /api/assessments/quiz/submit/ → Submit quiz answers
704 GET     /api/assessments/grades/      → Student's all grades
705 POST    /api/assessments/assignment/submit/ → Upload assignment
```

```
706 PATCH /api/assessments/grade/ → Grade assignment (instructor)
707
708 FEEDBACK ENDPOINTS:
709 POST /api/feedback/review/ → Add course review
710 GET /api/feedback/reviews/{id}/ → Get course reviews
711 POST /api/feedback/announcement/ → Post announcement (instructor)
712
713 =====
714 DAILY RULES TO FOLLOW
715 =====
716
717 1. Always activate venv before any Django command:
718     venv\Scripts\activate
719
720 2. Always check for (venv) prefix in terminal before running Python
721
722 3. Commit to GitHub at least once every day
723
724 4. Test each API endpoint before building the frontend for it
725
726 5. Keep two terminal tabs open at all times:
727     Tab 1: Django server (python manage.py runserver)
728     Tab 2: React server (npm start)
729
730 6. Never commit the .env file to GitHub
731
732 7. If you get an error – read the last line first, it tells you what broke
733
734 =====
735         END OF 30-DAY PLAN
736         Learning Hub – Online Learning Platform
737 =====
738
```